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AUTOMATION AND THE MEANING OF WORK

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Replaceable but Employed: Automation and the Meaning of Work

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ABSTRACT

Can automation harm workers without replacing them? We study jobs in which workers value both producing useful output and knowing that the output depends on their own contribution. A credible machine alternative can weaken that second source of meaning even when the firm retains the worker. Our model shows that this loss raises compensation when wages adjust fully; when they adjust only partly, workers bear some of the loss themselves. It can also make automation more likely. An external developer may profit by publicly demonstrating a machine before licensing it, because the demonstration lowers the value of the human alternative. This "meaning externality" can create demand for the machine and make profitable development socially harmful. Better technical quality and greater public salience have different effects: quality improves output, while salience alone weakens human work. Automation can, therefore, reduce the value of work before it eliminates jobs.

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1 Introduction

Stevenson (2026) asks what would remain of purpose in a post-work economy. Income, she argues, is not the issue: the income gains from transformative artificial intelligence cannot be counted on to preserve meaning, pride, or purpose. She argues that what is at stake is captured by the Japanese notion of *ikigai*; literally, “that which makes life worth living.” *Ikigai* need not be one’s job: it spans family roles, community ties, hobbies, faith, and craft, and in survey data it tracks income far more weakly than life satisfaction does. Yet for many workers today, market work is a principal source of it. A job supplies structure, identity, and the sense of being responsible for something of value, alongside a wage. Stevenson’s concern is what replaces that source when jobs disappear. We ask a prior question: can automation drain a job of its meaning while the worker keeps it?

This possibility arises well before automation eliminates a job. A physician may continue diagnosing patients after an artificial intelligence system can do so, and a software engineer may continue writing code after a model can produce a functional implementation. Translators, designers, accountants, teachers, and analysts may likewise remain employed and may even outperform automated alternatives. Yet once a machine is seen as capable of the task, the work may be understood differently. What once signalled scarce competence, responsibility, or a necessary human contribution may come to look like selection, supervision, or confirmation.

We study this change in how work is valued. Specifically, we ask how automation affects labour markets and welfare when workers derive meaning not only from performing socially valuable tasks but also from believing that some of that value depends on their own contribution.

This question is separate from, although connected to, the standard question of whether automation displaces labour. Task-based theories of technological change study how automation reallocates tasks between labour and capital, changes labour demand, and affects productivity or factor shares (Autor et al., 2003; Acemoglu and Restrepo, 2018, 2019; Acemoglu, 2025). We study a *pre-displacement margin* that can operate before any such reallocation. The availability of an automated method can change the amenity value of a task that remains assigned to labour. Put another way, a worker can be *replaceable but employed*.

The key distinction is between usefulness and attributable contribution. Suppose that a human-produced diagnosis, translation, or lesson continues to help its recipient. If meaning depends only on the realised value of output, a technologically feasible substitute need not diminish it. By contrast, if meaning also depends on the value that arises from

the worker personally performing the task, an automated alternative changes the task even when it is not used. It improves the relevant counterfactual and thereby reduces the value attributed to human performance.

We formalise this idea in a single-task model. A firm can produce by employing a worker or, when a technological capability objectively exists, by using an automated system. Separately, the capability may be more or less salient and credible to the worker. This distinction matters because feasibility governs what the firm can deploy, while salience governs the fallback against which the worker interprets personal contribution. The worker obtains a meaning-related amenity from two sources. The first is a *useful-output effect*: producing socially valuable output is meaningful. The second is an *attributable-contribution effect*: it matters that some valuable output exists because the worker personally performs the task. A salient automation capability improves the perceived fallback available without the worker and reduces the second component.

Our first result isolates this pre-displacement effect. When the task remains allocated to labour, a salient capability reduces the amenity value of human performance. In the posted-wage benchmark, compensation adjusts fully to the worker's reservation wage, so the firm bears the loss through higher compensation and the retained worker's net utility is unchanged. We then allow partial wage adjustment. A retained worker bears the unadjusted share of the amenity loss, the firm bears the adjusted share, and total surplus falls by the full amount. This makes clear when automation literally harms a worker it does not replace and when it instead raises the cost of retaining that worker.

Our next result shows that devaluation can change adoption itself. Automation is governed by two effects. The *production effect* compares what a human and an automated system can produce net of their costs. The *meaning effect* captures the change in the amenity value of human work once that work becomes replaceable. With full wage adjustment, the meaning loss raises the cost of staffing the human task and can move the adoption boundary. Under partial adjustment, the same feedback is scaled by the share of the loss passed through to wages. Thus, worker harm and the adoption feedback need not move one-for-one.

In the strongest case, retained human performance becomes a meaning-related disamenity. The capability can then make a formerly meaningful task more vulnerable to automation than a task with the same productive characteristics and ordinary burdens but no meaning-related amenity or disamenity.¹

¹This targeting reversal is stronger than the basic devaluation result. Appendix E shows that devaluation, adoption feedback, and the development wedge survive when attributable contribution is bounded below, whereas the reversal relative to a neutral task requires signed meaning to become negative.

The development margin reveals an important distinction. A firm developing a machine for its own use internalises the loss of meaning because it anticipates the resulting increase in the cost of retaining labour. By contrast, an external developer that makes a capability salient before licensing a machine sells against a human alternative that the demonstration has already devalued, without bearing the associated amenity loss. We refer to this as a *meaning externality*. With partial wage adjustment, the developer monetises only the share of the loss that enters wages; the remainder is borne by retained workers and priced by no one. For otherwise identical machines and development costs, the externality can make privately profitable development socially harmful and, once meaning becomes negative, can favour formerly meaningful tasks over neutral ones.

The demonstration that generates this externality is itself a choice. When public disclosure is costless, deployment access remains excludable, and the affected firm cannot purchase enforceable secrecy before disclosure, the developer prefers public demonstration whenever it supports a positive licence fee. There is then a region in which the demonstration creates the market: a machine that could not be licensed against undevalued human work becomes licensable once its own demonstration has devalued that work.²

The externality also operates on intensive margins, and the distinction between actual quality and salience is important. A marginal improvement in technical quality raises automated output and can further erode the human alternative. By contrast, a pure increase in salience leaves deployment value unchanged but can still increase the licence fee by weakening the human alternative. The model, therefore, separates the ordinary return to better technology from a rent-shifting return to publicity, benchmark performance, or other credibility-enhancing investments.

We next embed the mechanism in a short-run occupational-sorting model in which workers differ in how much they value meaningful work. With a fixed number of retained positions, a fall in the meaning of exposed work raises the compensating wage differential required to staff it. When the number of exposed jobs also changes, two effects oppose one another: the *amenity-loss effect* raises the required wage premium, while the *job-quantity effect* changes the composition of workers who remain in exposed employment. The observed wage effect is therefore ambiguous, and wage evidence alone cannot identify whether meaning has changed.

Finally, we allow a job to consist of multiple tasks. This extension clarifies that the model is not a general presumption against labour-saving technology. Automation can

²With a single affected firm and complete contracting over secrecy, a payment for nondisclosure can weakly dominate public demonstration in that region. Public disclosure is more robust when the harm is dispersed across many firms or workers who cannot coordinate to buy silence.

improve a job by removing burdensome tasks or creating time for more valuable human work. It can diminish a job when it removes or devalues the tasks that supplied judgment, authorship, or responsibility. Employment alone is therefore an incomplete measure of how technological change affects the quality of work.

Our analysis builds on the literature on meaningful work and compensating differentials. Cassar and Meier (2018) describe how mission, competence, and relatedness can enter labour supply and organisational design, Rosso et al. (2010) review the broader organisational literature on the sources of meaning at work, and Rosen (1986), Stern (2004), and Kesternich et al. (2021) show why valued job attributes affect wages and worker choice. We add a technological margin: automation changes an amenity of retained work by changing the counterfactual against which a human contribution is understood. The paper also connects to models of identity, motivated agents, and status (Akerlof and Kranton, 2000, 2005; Besley and Ghatak, 2005; Bénabou and Tirole, 2006; Besley and Ghatak, 2008). Here, the relevant change in interpretation is generated by a machine capability rather than by a competing worker, incentive scheme, or pay policy.

A companion analysis studies a different interaction between task amenities and automation. Gans (2026) considers O-ring jobs in which intrinsic task enjoyment can lead workers to supply productive time beyond their paid assignment, changing contracting, automation choices, and the informational content of payroll measures. Here, by contrast, the task amenity becomes endogenous to technological capability itself. The central margin is a pre-displacement devaluation of work that remains assigned to labour.

The distinction between income and purpose that motivates the paper has empirical content: income growth reliably raises evaluative wellbeing (Stevenson and Wolfers, 2008; Sacks et al., 2012), yet reported meaning in life does not track national income (Oishi and Diener, 2014), and sense of purpose responds to the characteristics of work and to transitions out of it (Weston et al., 2021; Yemiscigil et al., 2021). Relative to Stevenson (2026), we study a narrower market-work channel: a post-work economy is not required for work to lose some of its amenity value, since the loss can arise while workers remain employed, paid, and productive. Evidence that robotisation is associated with lower perceived meaningfulness and autonomy makes this margin plausible (Nikolova et al., 2024); our model isolates an effect that may arise before deployment is realised.

The rest of the paper is organised as follows. Section 2 introduces objective feasibility, worker salience, and the baseline model of meaning. Section 3 characterises retained-work devaluation and its incidence, adoption, external disclosure and development, the separate returns to quality and salience, and welfare. Section 4 studies occupational sorting and compensating wage differentials. Section 5 discusses jobs as task portfolios. Section 6

sets out empirical implications. Section 7 concludes. The appendices contain robustness exercises, extensions, and proofs.

2 Baseline Environment

We first consider a single market task. This stripped-down environment makes it possible to separate an automation capability from its adoption and, equally importantly, to separate the capability’s objective feasibility from its salience to the worker. A firm has an opportunity to produce one unit of a task-specific output. The task may be a legal analysis, a diagnostic assessment, a translation, a piece of code, or a design. A worker can perform the task. An automated system can perform it only if the relevant capability objectively exists. Table 1 collects the notation used in the main text.

2.1 Production, information, and timing

Human performance produces value $y > 0$. The unit task requires a fixed amount of worker time. A worker who performs it gives up leisure or another outside activity with value $c > 0$. The task also carries a direct burden $e \geq 0$, such as fatigue, boredom, stress, or unpleasant effort. These costs are conceptually distinct. The opportunity cost c arises because work uses time that could be allocated elsewhere. The direct burden e arises from what the worker must do during that time. An automated system, when objectively feasible, produces value $a \geq 0$ at deployment cost $k \geq 0$. Its net production value is

$$A \equiv a - k. \tag{1}$$

The deployment cost includes verification, integration, liability, and any other firm costs associated with using the automated system.

Assumption 1 (Captured task surplus). *The private production values y and a equal the total task surplus generated by human and automated output, respectively, and that surplus is captured by the firm. There is no additional customer surplus, market-power wedge, or output externality outside these objects.*

Assumption 1 is a benchmark restriction, not a claim that every product market has this property. It keeps the firm’s production problem and the welfare comparison on the same payoff scale.³

³Appendix E separates private and social output values and shows how the ordinary product-market wedge enters alongside the meaning externality.

Table 1: Notation

Symbol	Description
<i>Production and technology (Section 2)</i>	
y	Total task surplus from human-produced output, captured by the firm
c	Worker's opportunity cost of time
e	Direct task burden
a	Quality-adjusted total task surplus from automated output, captured by the firm
k	Deployment cost of the automated system
$A \equiv a - k$	Net production value of automation
$x \in \{0, 1\}$	Objective feasibility of the automated capability
$s \in [0, 1]$	Worker-perceived salience or credibility of that capability
<i>Meaning and incidence (Sections 2–3)</i>	
v	Perceived social value of human-produced output
$b_0; b(s, a)$	Initial perceived fallback; fallback at salience s
$\chi(a)$	Quality-dependent improvement in the perceived fallback
$\Delta_0 \equiv v - b_0$	Initial attributable contribution
$\alpha; \beta$	Weights on useful output and attributable contribution
$m(s, a)$	Meaning-related value of personal performance
θ	Worker's taste for meaningful work
$w^R(s, a)$	Reservation wage
$S_H(s; a)$	Human-production surplus at salience s
$L(a) \equiv \theta\beta\chi(a)$	Full-salience loss of human surplus
$\omega; \rho$	Pre-shock worker rent; wage pass-through of the meaning loss
<i>Development and welfare (Section 3)</i>	
$S_N \equiv y - c - e$	Surplus from the neutral task
$\bar{p}_M(a, k, s)$	Meaningful-task licence-fee cap at salience s
$\bar{p}_M^D(a, k); \bar{p}_M^P(a, k)$	Public-demonstration and private-offer fee caps
$\bar{p}_N(a, k)$	Licence-fee cap for the neutral task
R	Task-specific development cost
<i>Occupational sorting (Section 4)</i>	
$F; f$	Distribution and density of θ
$P; E$	Protected and exposed occupations
$n_P, n_E; w_P, w_E; m_P, m_E$	Position masses, wages, and meaning amenities
θ^*	Type of worker indifferent between occupations

We distinguish two states that a single capability indicator would otherwise conflate. Let $x \in \{0, 1\}$ denote *objective feasibility*. If $x = 1$, a machine capable of producing a exists and can be deployed; if $x = 0$, it does not. Let $s \in [0, 1]$ denote *salience*. The value $s = 1$ means that the worker regards the capability as fully credible and relevant to the task, $s = 0$ means that it has not changed the perceived fallback, and intermediate values capture partial salience or credibility. We impose $s \leq x$, so salience in the model is generated by a genuine capability rather than a false claim.

The ordinary public-capability comparison is between $(x, s) = (0, 0)$ and $(x, s) = (1, 1)$. Section 3.3 also uses the private state $(x, s) = (1, 0)$: the technology objectively exists and can be offered to the firm, but an unobserved rejected offer does not change the worker's perceived fallback. Section 3.4 permits interior salience.

The baseline timing is as follows. First, objective feasibility x is realised. Second, credible information or disclosure determines salience s . Third, the worker evaluates the meaning of personally performing the task. Fourth, the firm offers a wage for human labour or deploys automation when $x = 1$. Finally, production occurs.

For the posted-wage benchmark, the firm makes a take-it-or-leave-it offer to a worker whose outside utility is normalised to zero. The wage therefore equals the reservation wage. This makes the firm's production decision internalise the worker's experienced amenity through compensation. Proposition 2 relaxes full wage adjustment and separates worker and firm incidence without changing the aggregate amenity loss.

2.2 Useful output and attributable contribution

The worker obtains utility from wages and from the meaning of performing a market task. Meaning is distinct from the opportunity cost c and the direct task burden e . We distinguish two reasons why market work may be meaningful. First, the worker may value producing an output that helps others. Second, the worker may value the part of that outcome understood to exist because they personally perform the task.

Let $v > 0$ denote the social value that the worker attributes to the human-produced output. This is a perceived social value, not necessarily the market value y . For example, a physician may value the benefit of a diagnosis to a patient, while a teacher may value the benefit of a lesson to a student. Let $b(s, a)$ denote the value that the worker believes would be available without their personal performance. We refer to it as the *perceived fallback* and parameterise it as

$$b(s, a) = \underbrace{b_0}_{\text{initial fallback}} + \underbrace{s\chi(a)}_{\text{salience-induced improvement}}, \quad (2)$$

where $b_0 \geq 0$. Throughout, $\mathbb{R}_+ = [0, \infty)$, $\chi : \mathbb{R}_+ \rightarrow \mathbb{R}_+$, and $\chi(0) = 0$. Before a capability becomes salient, the fallback b_0 may include another worker, a less effective process, or non-production. A low-quality automated output changes little about the counterfactual; a high-quality output changes more.

The object $\chi(a)$ is measured in the same units as v and b . It is the perceived increase in fallback value produced by a fully salient capability of quality a . Technical quality and salience are nevertheless distinct: a changes what the machine can produce, while s changes how strongly that capability enters the worker's perceived comparison. This distinction permits a private capability to be objectively feasible without yet changing the meaning of retained work.

The worker's perceived attributable contribution is the gap $v - b(s, a)$. It is useful to name its initial level:

$$\Delta_0 \equiv v - b_0, \quad (3)$$

so that attributable contribution at salience s is $\Delta_0 - s\chi(a)$. Thus Δ_0 is not a primitive taste for human distinctiveness. It is the value that the worker initially understands to be attributable to personal performance.

The signed meaning-related value obtained by a worker who personally performs the task is

$$m(s, a) = \underbrace{\alpha v}_{\text{useful-output effect}} + \underbrace{\beta [\Delta_0 - s\chi(a)]}_{\text{attributable-contribution effect}}, \quad (4)$$

where $\alpha \geq 0$ and $\beta \geq 0$.

The useful-output effect captures the idea that it is meaningful to create an output that helps a customer, patient, student, client, or other user. The term αv is direct experienced amenity from understanding the output as socially useful, not a second entry for the recipient's surplus. Perceived usefulness is a distinct and measurable attribute of work: task significance affects effort and performance (Grant, 2008), and a non-trivial share of workers report regarding their own jobs as socially useless (Dur and van Lent, 2019). The attributable-contribution effect captures a different idea: it is meaningful to understand oneself as responsible for some of that value. A salient capability changes the fallback and, with it, the value attributed to human performance, even when human output and its perceived usefulness v are unchanged.⁴

The variable $m(s, a)$ is signed. A positive value is a meaning-related amenity; a negative

⁴For salience differences with v fixed, the attributable-contribution component aggregates into $\Delta_0 - s\chi(a)$. Levels of meaning do not reduce to that single index, because v also enters the useful-output term: tasks may differ in perceived social usefulness, in the initial fallback, and in how sharply a credible capability changes that fallback.

value is a meaning-related disamenity over and above the ordinary task burden e . When $b(s, a) > v$, the worker perceives personal performance not merely as less necessary but as producing less value than the salient fallback. This signed region generates the strongest reversal results.

Importantly, deployment cost k does not enter (2). This is the pure-capability version of the mechanism. It captures situations in which a machine's ability to perform a task changes the worker's perceived contribution even if implementation remains expensive. A translator may find that a task reflects less attributable expertise once an automated system can produce a reasonable translation, notwithstanding a particular employer's deployment arrangements. A physician may care that a diagnosis could increasingly be produced by a system even when liability or verification makes replacement unprofitable.⁵

A worker with taste for meaningful work $\theta > 0$ obtains $w - c - e + \theta m(s, a)$ from accepting the task at wage w . The reservation wage is therefore

$$w^R(s, a) = \underbrace{c}_{\text{opportunity cost}} + \underbrace{e}_{\text{direct burden}} - \underbrace{\theta m(s, a)}_{\text{meaning-related value}}. \quad (5)$$

That is, meaning lowers the wage required for participation when $m(s, a) > 0$ and raises it when $m(s, a) < 0$. For the baseline comparative static, c and e are fixed across salience states, which isolates the interpretation of retained human work rather than its duration or ordinary unpleasantness.⁶

Assumption 2 (Meaningful attributable contribution). $\theta > 0$, $\beta > 0$, $\Delta_0 > 0$, and $\chi(a) > 0$.

Assumption 2 says that the worker values meaningful market work, cares about attributable contribution, initially understands personal performance as adding value relative to the fallback, and regards a fully salient capability as improving that fallback. Since $v > 0$ and $\alpha \geq 0$, it implies $m(0, a) > 0$. We do not impose an upper bound on $\chi(a)$; a capability may merely reduce the amenity or may make retained human performance a disamenity.

Define the human-production surplus at salience s by

$$S_H(s; a) \equiv y - c - e + \theta m(s, a). \quad (6)$$

The quality argument is displayed because changes in a alter both automated value and

⁵Appendix B studies a more conservative, economically viable fallback, $b^{\text{ev}}(s, a, k) = b_0 + s\chi((a - k)_+)$ with $(u)_+ \equiv \max\{u, 0\}$, under which salience affects meaning only when automated production has positive net value, and derives the corresponding adoption threshold.

⁶The model summarises forgone leisure through c and does not include an intensive labour-supply choice.

the perceived fallback.

Assumption 3 (Viable paid human work). $c + e > \theta m(0, a)$ and $S_H(0; a) > 0$.

The first condition ensures a positive pre-capability reservation wage. The second ensures that human production is initially viable both socially and privately under Assumption 1.⁷

At the reservation wage, the firm's profit from human production is

$$y - w^R(s, a) = y - c - e + \theta m(s, a) = S_H(s; a).$$

The worker's amenity is not a free resource. It lowers the compensation required for participation, while a loss of meaning reduces the surplus available from human production. Given objective feasibility x and salience s , the firm's production choice therefore maximises

$$\max \left\{ \underbrace{S_H(s; a)}_{\text{human production}}, \underbrace{x A}_{\text{automated production}}, \underbrace{0}_{\text{non-production}} \right\}. \quad (7)$$

Equation (7) makes the information distinction explicit. Objective feasibility governs whether automation can be used; salience governs the value and compensation of retained human work. Under full wage adjustment, the employing firm internalises the worker's experienced amenity conditional on (x, s) . The upstream externality arises when a separate developer can change s before pricing access without bearing the resulting loss in S_H .

3 Capability, Adoption, Development, and Welfare

We now characterise the effect of a fully salient technological capability. We first consider retained human work and distinguish the aggregate amenity loss from its incidence. We then show how devaluation can move the adoption boundary. Finally, we study disclosure and development by an external developer, separate technical quality from pure salience, and evaluate welfare. Unless otherwise stated, the public-capability comparison is from $(x, s) = (0, 0)$ to $(x, s) = (1, 1)$.

⁷Standing conventions: the baseline propositions use a fixed quality a for which Assumption 2 holds, and when a result requires continuity, differentiability, or monotonicity of χ , it states that condition. Feasible production choices are human production, automation when $x = 1$, and non-production; strict adoption means strict payoff dominance over the other feasible choices. If alternatives tie, any tied choice may be selected and welfare statements use their common surplus value. A buyer offered a licence at its fee cap is taken to accept.

Assumption 4 (Experienced-utility welfare). *The term $\theta m(s, a)$ is direct experienced utility evaluated under the worker’s actual information state s . Wages and licence payments are transfers. Welfare is the sum of production surplus and experienced utility, net of real costs.*

Assumption 4 makes the normative criterion explicit.⁸ To keep the formal statements compact, Assumptions 1–4 are maintained throughout this section without further mention; the disclosure environment introduced in Section 3.3 adds one further assumption.

3.1 Retained human work and incidence

From (4), full salience changes human meaning by

$$m(1, a) - m(0, a) = -\beta\chi(a) < 0. \quad (8)$$

We call this *meaning devaluation*. It arises from the improved perceived fallback, not from deployment: the machine need never be used. Our first result records the consequences for retained work.

Proposition 1. *Suppose $S_H(1; a) > \max\{A, 0\}$, so that human production remains the firm’s unique choice before and after the capability becomes objectively feasible and fully salient. Then full salience lowers meaning by $\beta\chi(a)$, raises the reservation wage by $\theta\beta\chi(a)$, and lowers human-production surplus by $\theta\beta\chi(a)$.*

The worker performs exactly the same market task and produces the same output y . Nevertheless, the task is a worse amenity once the automated capability becomes salient. Under the posted-wage benchmark, the worker must be compensated through a higher wage in order to participate, and the firm bears the resulting loss of human-production surplus. The retained worker’s net utility remains at the normalised outside utility. Proposition 1 therefore establishes a lost amenity and its compensating variation, not a fall in a fully compensated worker’s realised utility.

The condition $S_H(1; a) > \max\{A, 0\}$ focuses attention on a task for which human production remains strictly better than automation and non-production. This special case distinguishes capability from adoption. When A is small because deployment is costly or automated output has limited value, an ordinary production model records no change. Here the meaning effect remains active even though the production effect favours continued human performance.

⁸It is not an ex ante preference over information, nor an “informed preference” criterion that replaces the worker’s experienced beliefs with an external evaluator’s beliefs. The welfare discussion in Section 3.5 returns to this distinction.

Define the full-salience loss of human surplus as

$$L(a) \equiv S_H(0; a) - S_H(1; a) = \theta\beta\chi(a) > 0. \quad (9)$$

The next result separates this aggregate loss from its incidence and shows how incidence governs the immediate adoption feedback. Suppose the worker enters with a wage $w_0 = w^R(0, a) + \omega$, embodying a pre-shock rent $\omega \in [0, S_H(0; a))$, and that the wage responds to full salience by $w_1 = w_0 + \rho L(a)$, where $\rho \in [0, 1]$ is the pass-through of the meaning loss into wages. Assume $\omega \geq (1 - \rho)L(a)$, so the post-salience contract remains acceptable to the worker.⁹

Proposition 2. *Under the wage-adjustment environment above: (i) if human production remains chosen, worker utility falls by $(1 - \rho)L(a)$, firm profit falls by $\rho L(a)$, and total surplus falls by $L(a)$; (ii) relative to the pre-salience contract, devaluation induces automation if and only if*

$$\max\{S_H(0; a) - \omega - \rho L(a), 0\} < A \leq S_H(0; a) - \omega. \quad (10)$$

Proposition 2 gives the worker-harm result. When $\rho = 1$, the firm bears the entire loss through compensation and the worker's utility is unchanged; setting $\omega = 0$ gives the exact posted-wage benchmark. When $\rho = 0$, a retained worker bears the entire loss, but there is no immediate wage-cost feedback into adoption. This is because none of the amenity loss enters the firm's labour cost. Intermediate pass-through divides the loss between worker and firm, while total welfare falls by $L(a)$.

3.2 Meaning-induced adoption

We now return to the posted-wage benchmark $(\rho, \omega) = (1, 0)$, in which human profit equals total human surplus. This benchmark isolates the maximum immediate feedback from devaluation into the firm's adoption decision.

Say that the capability produces *meaning-induced adoption* if automated production is strictly adopted after the capability becomes objectively feasible and fully salient, although it would not strictly dominate human production if the worker retained the pre-salience meaning of the task. The following proposition characterises exactly when this occurs.

⁹This retention condition allows pre-existing rents, switching costs, or an incumbent contract to keep the worker employed despite incomplete compensation. If it fails, worker exit or renegotiation must also be specified.

Proposition 3. *Meaning-induced adoption occurs if and only if*

$$\max\{S_H(1;a), 0\} < A \leq S_H(0;a). \quad (11)$$

The strict first inequality requires automation to beat both post-salience human production and non-production; the weak second says that, at the pre-salience value of human work, automation would not have strictly displaced labour. The ranking changes because salience has lowered the value of human work. Intuitively, in the absence of the meaning loss, human production would remain preferred because the amenity makes labour available at a lower wage; once the capability becomes salient, the wage required for human work rises and automation can become preferable. Proposition 2 shows that, with partial wage adjustment, the corresponding immediate feedback is $\rho L(a)$ rather than the full loss.¹⁰

To return to the motivating examples, a firm may initially prefer a human analysis, translation, design, or diagnosis even when an automated version is technically possible. If performing the task supplied a non-pecuniary benefit to skilled workers, that benefit was part of the reason human labour was available at a given wage. Once the task no longer carries the same meaning, the wage comparison changes. In this sense, a capability can first reduce the attractiveness of remaining human work and then replace it.

Proposition 3 requires only a loss of meaning. A stronger comparison asks whether the formerly meaningful task becomes more vulnerable than an otherwise identical task with no meaning-related amenity or disamenity—the *neutral task*. Let

$$S_N \equiv y - c - e \quad (12)$$

denote the surplus from this human-performed task, which shares the same productive characteristics, opportunity cost, and direct burden.

Corollary 1. *Suppose $S_N > 0$. An automated alternative with net value A strictly displaces the formerly meaningful task after full salience, but would not strictly displace the neutral task, if and only if $\max\{S_H(1;a), 0\} < A \leq S_N$. This interval is non-empty if and only if $m(1, a) < 0$.*

¹⁰In terms of deployment costs, substituting $A = a - k$ and imposing $k \geq 0$ shows that (11) corresponds to $k \in [\max\{0, a - S_H(0;a)\}, a - \max\{S_H(1;a), 0\}]$ whenever $a > \max\{S_H(1;a), 0\}$, and to an empty set otherwise. The interval's length equals $L(a)$ exactly when post-salience human production remains non-negative and $a \geq S_H(0;a)$, so that the lower bound is set by pre-salience human surplus rather than by $k \geq 0$.

Corollary 1 isolates an *amenity reversal*.¹¹ A formerly meaningful task becomes more vulnerable than an otherwise identical neutral task only when full salience turns human performance into an additional disamenity. Automation then does not merely remove a wage discount associated with meaningful work; it creates a compensating premium for continuing to perform the task.¹²

3.3 External development incentives

The adoption feedback raises an upstream question: does it change which task-specific capabilities an external developer chooses to create and disclose? Continue to assume $S_N > 0$. Throughout this subsection, the affected firm employs its worker in the posted-wage benchmark; we record below how partial wage adjustment scales the share of the loss a developer can monetise. A developer can create either the formerly meaningful-task machine or the neutral-task machine at task-specific development cost $R \geq 0$. Both machines have quality a , deployment cost k , and net automated value A , excluding any licence fee.

Assumption 5 (Disclosure and contracting environment). *After paying R , the developer chooses either a credible public demonstration, which sets $s = 1$ before licensing, or a private offer that remains unobserved by the worker if rejected and therefore leaves $s = 0$. Public demonstration is costless, does not destroy the excludability of deployment access, and is observed before the fee is set. The developer cannot contract with the affected firm before choosing disclosure and cannot commit to suppress truthful public disclosure through an enforceable secrecy agreement. The developer makes a take-it-or-leave-it fee offer, and the buyer accepts at its fee cap.*

Assumption 5 states the contracting restriction required for endogenous public demonstration; the boundary created by complete contracting over secrecy is derived below.

For any salience s , the highest meaningful-task licence fee consistent with adoption is

$$\bar{p}_M(a, k, s) = [A - \max\{S_H(s; a), 0\}]_+ . \quad (13)$$

The full public-demonstration, private-offer, and neutral-task caps are, respectively,

$$\bar{p}_M^D(a, k) \equiv \bar{p}_M(a, k, 1) = [A - \max\{S_H(1; a), 0\}]_+ , \quad (14)$$

¹¹The interval is stated in net-value space. Since $A = a - k$ with $k \geq 0$, a feasible reversal at fixed quality a additionally requires $a > \max\{S_H(1; a), 0\}$; negative post-salience meaning alone does not guarantee a machine capable enough to occupy the interval.

¹²Appendix E shows that bounding attributable contribution below at zero preserves devaluation and adoption feedback but rules out this stronger reversal.

$$\bar{p}_M^P(a, k) \equiv \bar{p}_M(a, k, 0) = [A - S_H(0; a)]_+, \quad \bar{p}_N(a, k) = [A - S_N]_+. \quad (15)$$

The private cap uses $S_H(0; a)$ because objective feasibility is $x = 1$, while a rejected offer leaves the worker's salience at $s = 0$. The neutral task has no meaning term, so its cap is independent of disclosure. The following proposition characterises the developer's disclosure choice.

Proposition 4. (i) $\bar{p}_M^D \geq \bar{p}_M^P$, with strict inequality if and only if $A > \max\{S_H(1; a), 0\}$, so public demonstration is weakly optimal and strictly optimal whenever the demonstrated licence commands a positive fee. (ii) If $\max\{S_H(1; a), 0\} < A \leq S_H(0; a)$, then $\bar{p}_M^P = 0 < \bar{p}_M^D$: a positive licence fee is supportable only after public demonstration. (iii) Under a private offer, development followed by adoption is strictly profitable if and only if it strictly raises welfare.

Under the stated timing, public disclosure worsens the outside option against which the licence is priced. Part (ii) is the market-creation region: the machine is not worth buying against undervalued human work, but becomes licensable after its own demonstration weakens that alternative. Part (iii) locates the externality. A genuinely private offer does not change the perceived fallback before the transaction and therefore aligns private and social development incentives in this benchmark.

The comparison requires that a private offer remain private. Where offers leak, where a demonstration to one firm is observed by workers elsewhere, or where deployment itself makes the capability salient across an occupation, the public cap is the relevant one. The result is consistent with a familiar pattern in markets for artificial-intelligence capabilities, in which public demonstrations and benchmark performances precede licensing. In the model, such demonstrations are not only marketing to buyers; they also weaken the human alternative against which buyers bargain.

The restriction on contracting over secrecy is not innocuous, and it is worth deriving the boundary it creates. Let $B \equiv \max\{S_H(1; a), 0\}$ and consider the market-creation region $B < A \leq S_H(0; a)$. Public demonstration and licensing yield the developer $A - B$ and leave the firm with payoff B . If a single firm could instead purchase an enforceable commitment to silence before disclosure, its willingness to pay to preserve undervalued human production would be $S_H(0; a) - B$. Since $S_H(0; a) - B - (A - B) = S_H(0; a) - A \geq 0$, a silence contract weakly dominates public demonstration, and strictly so when $A < S_H(0; a)$: under complete pre-disclosure contracting, the predicted transaction may be a payment for silence rather than a demonstration.¹³

¹³With N symmetric prospective licensees and no coordinating coalition, public licensing yields $N(A - B)$, and a unilateral silence purchase cannot block disclosure whenever $N(A - B) > S_H(0; a) - B$. Dispersed harm and free riding therefore provide one foundation for Assumption 5.

When public demonstration induces automation, so $A > \max\{S_H(1;a), 0\}$, the developer's licensing margin exceeds the gross social gain from replacing pre-salience human production by

$$\underbrace{\bar{p}_M^D}_{\text{developer's gross licensing margin}} - \underbrace{[A - S_H(0;a)]}_{\text{gross social gain}} = \underbrace{S_H(0;a) - \max\{S_H(1;a), 0\}}_{\text{meaning externality}} > 0. \quad (16)$$

If post-salience human production remains viable, the externality equals $L(a)$; otherwise non-production replaces human work as the buyer's outside option. Whenever $A > S_H(0;a)$, the same object is the developer's incremental revenue from choosing public demonstration over a private offer. In the market-creation region, the demonstration accounts for the entire licensing margin. In other words, the developer is paid for the damage its own disclosure does to the human alternative.

These caps price against a firm whose wage fully absorbs the meaning loss. Suppose instead, as in Proposition 2, that the worker holds a pre-shock rent ω , that the wage passes through the share ρ of the loss, and that the worker continues to staff the task. The firm's human-production payoff is then $S_H(0;a) - \omega$ under a private offer and $S_H(0;a) - \omega - \rho L(a)$ after public demonstration.¹⁴ The developer's incremental revenue from public demonstration is therefore at most $\rho L(a)$: only the share of devaluation that enters the wage bill weakens the outside option the developer prices against, while the total loss from full salience remains $L(a)$. The meaning externality thus splits into a monetised component, bounded by $\rho L(a)$, and an unpriced component $(1 - \rho)L(a)$ borne by the retained worker. At $\rho = 0$, public demonstration earns the developer nothing over a private offer yet destroys the amenity all the same—the harm is unchanged while the private return to generating it disappears—and market creation requires $\rho > 0$.

The next proposition asks which tasks the externality directs development towards.

Proposition 5. *Compare machines with the same a , k , and R , and suppose $S_N > 0$ and $m(1, a) < 0$. Then $\bar{p}_M^D \geq \bar{p}_N$, with strict inequality whenever $A > \max\{S_H(1;a), 0\}$. For any such A , there are development costs $R \in [\bar{p}_N, \bar{p}_M^D)$ for which development for the formerly meaningful task is profitable while development for the neutral task is not.*

If instead $m(1, a) > 0$, the ranking reverses, strictly whenever $A > S_N$; at $m(1, a) = 0$ the two caps coincide. The meaning externality always raises the public-channel licensing margin relative to a counterfactual in which the meaningful task retains its pre-salience amenity. But a formerly meaningful task becomes a better target than a neutral task only

¹⁴The corresponding fee caps are $[A - \max\{S_H(0;a) - \omega, 0\}]_+$ and $[A - \max\{S_H(0;a) - \omega - \rho L(a), 0\}]_+$.

after the amenity reverses sign—a *targeting reversal*. The upstream externality is therefore broader than its strongest targeting implication.

Corollary 2. *Suppose $A > \max\{S_H(1;a), 0\}$. Publicly demonstrated development is profitable while reducing welfare if and only if*

$$R \in [0, A - \max\{S_H(1;a), 0\}) \cap (A - S_H(0;a), \infty), \quad (17)$$

a set that is always non-empty.

Corollary 2 does not imply that external development is generally excessive. When the productive gain is sufficiently large, development raises welfare despite the loss of meaning. The result identifies costs for which the developer monetises a human alternative weakened by public disclosure while omitting that loss from its objective.¹⁵

3.4 Technical quality and salience

The externality concerns not only whether a capability is created and disclosed, but also how capable and how salient it becomes. Technical quality a raises net automated production value $A = a - k$ and, through $\chi(a)$, may improve the perceived fallback. Salience s , by contrast, changes how strongly a fixed capability enters that fallback without changing deployment value. The following proposition separates the two returns.

Proposition 6. *Suppose $A > \max\{S_H(s;a), 0\}$ and $S_H(s;a) > 0$, so that adoption occurs while human production remains the buyer’s outside option, and let χ be differentiable at a with $\chi'(a) > 0$. Conditional on adoption, a marginal improvement in quality raises social surplus and the neutral-task cap one-for-one, but raises the meaningful-task cap by $1 + s\theta\beta\chi'(a)$; a marginal increase in salience leaves social surplus unchanged, but raises the meaningful-task cap by $\theta\beta\chi(a)$.*

The quality result separates the production and devaluation channels: a better machine creates one unit of ordinary production value per unit of a , but an external seller can also capture the reduction in a viable human alternative. The salience result is not a quality result relabelled in prose. Publicity, benchmark performance, or other credibility-enhancing activity can raise the fee without raising deployment value.¹⁶ It follows that if salience costs the developer $\kappa(s)$, with $\kappa'(s) > 0$, the developer’s marginal net return

¹⁵Under a private offer, the interval disappears by Proposition 4(iii). Under complete confidentiality contracting, the analysis above shows that the predicted transaction may instead be a payment for silence.

¹⁶The derivatives hold k and R fixed, with $s \in (0, 1)$ for the salience derivative. Once $S_H(s;a) < 0$, non-production replaces human work as the buyer’s outside option: both caps then move one-for-one with quality and are insensitive to salience, so further devaluation earns the developer nothing.

to salience is $\theta\beta\chi(a) - \kappa'(s)$ while the planner's conditional marginal return is $-\kappa'(s)$: costly salience can be privately valuable even when, conditional on adoption, it creates no production benefit.

For comparison, integration removes this rent-shifting return. A firm developing a machine for its own use values an improvement in quality through production, while any reduction in the amenity of retained labour raises its own compensation cost.¹⁷

3.5 Welfare

We now evaluate a fully salient capability once it exists, regardless of how it was developed. By Assumptions 1 and 4, wages and licence fees are transfers, while y , a , worker costs, and experienced meaning enter welfare. Before the capability exists, welfare is $S_H(0; a) > 0$; after objective feasibility and full salience, welfare is $\max\{S_H(1; a), A, 0\}$, evaluated at the firm's choice.

Proposition 7. *Relative to the pre-capability economy, a fully salient capability changes welfare by $-L(a)$ when human production remains weakly optimal, by $A - S_H(0; a)$ when automation is weakly optimal, and by $-S_H(0; a)$ when non-production is weakly optimal.*

The first case is the pre-adoption cost: there need be no change in output or employment for the capability to lower welfare. Full wage compensation changes the incidence of the loss; it does not recreate the destroyed amenity. The second case includes meaning-induced adoption. When $\max\{S_H(1; a), 0\} < A < S_H(0; a)$, the capability induces automation and lowers welfare by $S_H(0; a) - A$; the change turns positive only for $A > S_H(0; a)$. This produces a form of so-so automation (Acemoglu and Restrepo, 2019) through a distinct channel: the capability itself destroys an amenity that had made human production valuable. The third case records the outside option of non-production.

The welfare criterion counts belief-dependent meaning as experienced utility, so a truthful demonstration can lower welfare by changing a valued belief about attributable contribution. This is a substantive normative assumption, not a theorem about all possible criteria; the paper adopts experienced utility because the mechanism concerns the quality of work as workers actually experience it.¹⁸

¹⁷Appendix D.3 formalises this comparison and distinguishes a salient internal capability from one kept confidential from the worker.

¹⁸Belief-dependent utility is a recognised object in welfare analysis (Caplin and Leahy, 2001), and workers may rationally avoid information about a capability (Golman et al., 2017). An ex ante information-preference criterion could value or dislike learning itself; an informed-preference criterion could refuse to count utility sustained by mistaken beliefs. Either could rank disclosure differently. Counting experienced meaning does not imply that policy should preserve inefficient tasks or suppress truthful information; it means only that output gains and changes in job amenities are distinct entries in the stated criterion.

Proposition 7 should not be read as a claim that the state should prohibit a capability whenever it diminishes meaning. The model has one task and omits downstream innovation benefits, option value, non-market sources of purpose, and consumer surplus beyond the value included in y and a . The result is narrower: under the stated surplus criterion, meaning is a real term in welfare accounting and cannot be assumed to be dominated by a small technological gain.

4 Occupational Sorting and Wage Differentials

The baseline model uses a single worker and take-it-or-leave-it wage setting. This section embeds the mechanism in a competitive sorting environment to establish two implications: devaluation of retained work is a labour-supply shock that raises a compensating differential, and a large enough shock reverses which workers select into exposed work. The object is a relative-wage equilibrium for given masses of positions, not a full model of vacancy creation, product demand, or absolute wage levels.

There is a unit mass of workers indexed by their valuation of meaningful work θ , distributed with continuously differentiable distribution F and strictly positive density f on $[\underline{\theta}, \bar{\theta}] \subset \mathbb{R}_+$. All workers are employed in one of two human-performed occupations: occupation P is protected from the capability shock and occupation E is exposed. Position masses $n_P, n_E \in (0, 1)$ sum to one and are fixed, representing the short run in which firms retain human positions but wages adjust. A worker of type θ receives $w_o + \theta m_o$ in occupation $o \in \{P, E\}$, where w_o is the wage and m_o the meaning amenity, and chooses the larger.¹⁹

Suppose protected work supplies the greater amenity, $m_P > m_E$, as it will after a sufficiently large shock to exposed work. Workers with high θ then select into protected work, the indifferent type is $\theta^* = F^{-1}(n_E)$, and the indifference condition pins down the familiar compensating differential

$$w_E - w_P = \theta^*(m_P - m_E). \quad (18)$$

The exposed occupation pays a premium that compensates the marginal worker for its lower meaning amenity.

¹⁹Opportunity costs and direct burdens are normalised equal across occupations; fixed differences would add a constant to the differential without changing the comparative static. In equilibrium, almost every worker chooses an occupation maximising $w_o + \theta m_o$ and the mass assigned to each occupation equals its position mass. Because position masses are given and there is no firm-side zero-profit condition, only $w_E - w_P$ is determined; a numeraire or demand condition would pin down wage levels.

A full-salience shock to a single exposed task maps from the baseline as $m_E^1 - m_E^0 = -\beta\chi(a)$; more generally, let a shock reduce exposed meaning by $dm_E < 0$ without altering the number of retained positions. With positions fixed, the cutoff θ^* is fixed, and differentiating (18) gives

$$d(w_E - w_P) = -\theta^* dm_E > 0. \quad (19)$$

Meaning devaluation is thus a labour-supply shock for retained jobs. If an automated capability makes exposed work less meaningful while the required number of human workers is unchanged, exposed work must pay more. This is the wage counterpart of Proposition 1.

The fixed-quantity assumption deliberately shuts down displacement. To display the composition effect, suppose instead that every reduction in exposed positions is matched one-for-one by an increase in protected positions, so $n_P = 1 - n_E$ and all workers remain employed. If n_E changes while m_P is fixed, differentiating (18) gives

$$d(w_E - w_P) = \underbrace{-\theta^* dm_E}_{\text{amenity-loss effect}} + \underbrace{\frac{m_P - m_E}{f(\theta^*)} dn_E}_{\text{job-quantity effect}}. \quad (20)$$

When automation reduces the meaning of retained work, the first term is positive. When it also reduces exposed positions, so $dn_E < 0$, the second term is negative: the workers remaining in exposed employment are drawn from a smaller, lower- θ set and require a smaller premium. Wage evidence alone therefore cannot identify whether meaning has changed. A rising relative wage for retained exposed workers is consistent with the amenity-loss effect dominating; a falling one is consistent with displacement or composition dominating. The relevant comparison examines wages jointly with job quantities and worker sorting.²⁰

A large enough shock also reverses sorting. Let m_E^0 and m_E^1 denote the pre- and post-shock meaning amenities of exposed work, with m_P unchanged, and suppose

$$m_E^0 > m_P > m_E^1, \quad (21)$$

so exposed work is initially the higher-meaning occupation and becomes the lower-meaning one. The following proposition records the resulting *sorting reversal*.

Proposition 8. *Suppose all workers remain employed in the two occupations, position masses are*

²⁰Equation (20) is a comparative-static accounting formula under the stated one-for-one reallocation, not a general labour-market equilibrium. If displaced workers instead become unemployed, leave the market, or enter other occupations, additional margins must be modelled.

fixed, and (21) holds. In the pre-shock equilibrium, exposed work attracts the upper tail of the θ distribution, with cutoff $\theta_0^ = F^{-1}(n_P)$; in the post-shock equilibrium, it attracts the lower tail, with cutoff $\theta_1^* = F^{-1}(n_E)$. Workers with $\theta > \max\{\theta_0^*, \theta_1^*\}$ choose exposed work before the shock and protected work after it, while workers with $\theta < \min\{\theta_0^*, \theta_1^*\}$ do the reverse.*

Proposition 8 compares two static sorting equilibria. A profession can retain substantial labour demand while becoming less attractive to precisely those who would formerly have selected into it because it supplied purpose, craft, or responsibility. The distinction from a skill-biased or task-displacement account is that a worker can avoid an occupation even when monetary prospects are protected, because the work is no longer the same amenity.²¹

5 Jobs as Task Portfolios

The preceding sections treat an occupation as a single task. Actual jobs combine many tasks, and automation may affect them differently. A physician spends time diagnosing, communicating, recording information, and coordinating care; a software engineer designs systems, writes code, checks output, and debugs failures. Whether automation improves or diminishes meaningful work depends on which parts of these jobs it changes. Appendix C states a task-level accounting identity that decomposes the change in an occupation's non-pecuniary value into three terms: a retained-task term, in which tasks stay with labour but their salience, burden, or time weight changes; a removed-task term for tasks no longer performed by labour; and a new-task term for human tasks that technological change creates.

The decomposition is an identity rather than a behavioural theorem, but it disciplines both optimistic and pessimistic readings of the mechanism. Automation can improve a job. If a removed task was burdensome—administrative work, repetitive data entry, physically unpleasant requirements—its removal deletes an existing disamenity, and freed time can be reallocated towards high-meaning human tasks. Automation can instead diminish a job when it removes or devalues the tasks that supplied judgment, craft, authorship, or responsibility while leaving workers to monitor outputs and address exceptions. The retained-task term contains the baseline mechanism: a task can stay with labour and become less meaningful because a salient capability improves its perceived fallback, and if the devaluation is large enough, a central task can move from amenity to disamenity

²¹Interpreting the comparison as changes in training, entry, or occupational composition requires enough mobility for workers to re-sort; the transition dynamics are not modelled. In a shorter run with occupation-specific capital, the same shock may instead appear as dissatisfaction, quits, retention pressure, or wage renegotiation among incumbents.

before deployment removes it. Employment alone is therefore an insufficient measure of how technological change affects the quality of work; the set, intensity, and interpretation of retained tasks matter.²²

The same distinction applies to technologies that assist rather than replace. A tool can raise output while reducing the share of the outcome that the worker attributes to their own judgment. A decision-support system that enables a professional to exercise better judgment may preserve or raise meaning; a system that generates an answer for formal approval may leave the worker more productive but less responsible for the valuable output. This margin supplies one microfoundation for the design choice that Brynjolfsson (2022) identifies between human-like substitution and augmentation: two technologies with the same measured output can differ in the share of the outcome that the worker understands as their own. Appendix D formalises the comparison and states an integrated-development benchmark in which a firm creating a capability for its own use internalises the meaning loss through its wage bill, to the extent that wages adjust.

6 Empirical Implications and Interpretation

The analysis is theoretical, but it identifies patterns that empirical work may seek to distinguish. The central measurement challenge is to separate objective feasibility from worker salience, salience from adoption, and retained-work devaluation from displacement. The model points to combinations of wage, quantity, sorting, and direct amenity measures rather than a single sufficient statistic.

6.1 Wages, quantities, and sorting

The most direct implication concerns tasks that remain performed by labour after an automated capability becomes salient. Equation (19) implies that a fall in the meaning amenity shifts labour supply away from exposed retained work. Conditional on a stable quantity of human positions, its relative wage should rise. The effect is likely to be most visible in occupations whose workers previously accepted lower compensation in exchange for authorship, mission, craft, or responsibility (Stern, 2004; Cassar and Meier, 2018; Kesternich et al., 2021; De Schouwer et al., 2025). For these occupations, salience should narrow an existing wage discount. If retained human performance becomes a disamenity, the discount can become a compensating premium.

²²The decomposition also clarifies why a job can become less meaningful even as its measured skill level rises: remaining workers may handle exceptional cases requiring considerable formal skill yet feel less responsible for the normal production of value.

This is not a prediction that workers in highly exposed occupations generally receive wage gains. A reduction in exposed positions activates the composition term in (20), and a richer production model could add direct labour-demand effects. Partial wage adjustment also matters: Proposition 2 predicts a smaller immediate wage response and a larger utility loss for incumbents when $\rho < 1$. Empirical work should therefore distinguish retained tasks, displaced tasks, and recomposed jobs, and should examine wages jointly with quits, vacancies, retention, job satisfaction, and employment quantities.

The sorting implication is similarly conditional. Workers who place a high value on meaning should move towards tasks or employers where responsibility remains more clearly human. With long training lags or occupation-specific capital, the short-run manifestations may instead be dissatisfaction, reduced effort, quits, or demands for compensation.²³

6.2 Measuring the pre-displacement margin

Because

$$m(s, a) = (\alpha + \beta)v - \beta b(s, a),$$

a single meaningfulness index does not identify perceived usefulness, attributable contribution, and the fallback separately. Surveys could ask whether workers believe their output helps others, whether a comparable outcome would be available without their personal performance, how credible and salient a particular capability is, and what compensation they would require to continue the unchanged task. Stated-preference methods that price job amenities directly (Mas and Pallais, 2017; Maestas et al., 2023) are well suited to the last object.

Occupational and task-level measures of artificial-intelligence exposure (Felten et al., 2021; Eloundou et al., 2024) proxy for tasks whose objective or perceived fallback may change. Interacting such exposure with the timing of a prominent public demonstration can help identify the pre-displacement margin because public salience may move before firm-level deployment. It does not, by itself, isolate the mechanism. The same event may alter expected future displacement, perceived job security, product demand, occupational status, bargaining positions, work organisation, or unmeasured informal use. A persuasive design would therefore combine exposure and event timing with evidence that the task remains human-performed, controls or measures for expected displacement, and direct

²³Selection should not be interpreted as mechanically stabilising measured meaning: if a survey measures the task amenity m_E , the fall remains visible among stayers, whereas if it measures experienced amenity θm_E , the exit of high- θ workers can alter the composition of responses.

measures of usefulness, attributable contribution, and perceived fallback.

This distinction matters for interpreting existing evidence. Nikolova et al. (2024) find that robotisation is associated with lower work meaningfulness and autonomy. Their evidence naturally combines task reallocation, workplace redesign, and the experience of working alongside deployed automation. The mechanism here is narrower: a credible public capability can change the interpretation of retained work before realised use. Generative artificial intelligence is a natural setting because public demonstrations can precede incorporation into firm workflows, but the interval between demonstration and deployment is informative rather than automatically causal.

The timing of adoption supplies a further implication. Propositions 2 and 3 imply that adoption may be preceded by deterioration in recruitment, retention, job satisfaction, or the wage bargain for a human-performed task. The machine need not become cheaper between those events. Instead, the task may become harder to staff at its previous wage because it no longer supplies the same amenity. The extent to which this labour-supply change immediately moves adoption depends on wage pass-through ρ .

The development predictions are conditional on the contracting environment. Under Assumption 5, public demonstration should precede licensing when devaluation supports a higher fee, and Proposition 6 identifies a wedge between the private and social marginal returns to quality and salience while human work remains a viable outside option—an incentive towards overinvestment, conditional on adoption, rather than a prediction about realised investment levels. The monetisable wedge rises with wage pass-through. With enforceable confidentiality contracts, the model instead predicts possible payments for nondisclosure (Section 3.3), so evidence on pre-disclosure contracts, benchmark publicity, the number of affected firms, and who captures licensing revenue is needed to distinguish the channels.

6.3 Interpretation and limits

The model does not equate meaning with indispensability. A valuable output does not cease to be useful merely because a machine can also produce it; this is why (4) has a useful-output term that is invariant to salience. A worker who values contribution only through realised output has $\beta = 0$ and experiences no benchmark devaluation.

The signed contribution term permits retained performance to become a disamenity. That assumption is necessary for the strongest comparison with a neutral task, but not for the basic mechanism; Appendix E shows that a bounded contribution function preserves devaluation, adoption feedback, and the development wedge while eliminating amenity

reversal. Empirical claims about targeting formerly meaningful work should therefore be tied to evidence of futility, obstruction, or another genuinely negative experience, not merely a decline in purpose.

Incidence is institution-dependent. The posted-wage benchmark has $\rho = 1$ and $\omega = 0$, so the firm bears the loss and the retained worker is fully compensated. Sticky wages, incumbent rents, bargaining frictions, occupation-specific capital, or monopsony can produce $\rho < 1$, in which case Proposition 2 gives a direct utility loss for retained workers and a weaker immediate adoption feedback. Later effects may still operate through quits, reduced effort, recruitment, or renegotiation, but those dynamics require additional structure.

Not every meaning loss is an externality. An employing firm that develops a machine for its own use internalises the effect on its human-production alternative to the extent that wages adjust—fully in the posted-wage benchmark, and only the share $\rho L(a)$ under partial pass-through, with the remainder falling on its own retained worker. The externality proper arises when a separate developer changes salience before pricing access and cannot be made to internalise the resulting loss through contracting.²⁴

Finally, the paper restricts attention to meaning obtained from market work. Family roles, community, friendship, care, and civic participation can supply purpose when paid work changes or disappears (Stevenson, 2026). Allowing workers to reallocate time and identity towards those activities is a natural extension. The present analysis establishes an earlier margin: the value of continuing paid work can fall before its hours or income disappear.

7 Conclusion

Automation can affect work without immediately replacing a worker. A task that remains assigned to labour can become a different amenity once an automated alternative is objectively feasible and salient. Separating feasibility from salience clarifies that the firm may be able to deploy a machine even when the worker does not yet incorporate it into the counterfactual against which personal contribution is understood.

In the posted-wage benchmark, salience reduces human-production surplus and raises the reservation wage while leaving a retained worker fully compensated. With partial wage adjustment and pre-existing rents, the same loss is divided between lower worker

²⁴The parties on the other side of disclosure have the opposite interest in salience: an employing firm benefits from an amenity that lowers its wage bill, while a worker may prefer not to learn about a capability that diminishes experienced meaning.

utility and lower firm profit. The share passed through to wages governs how strongly devaluation feeds back into adoption. Thus worker harm and the adoption channel are related but distinct.

A separate developer can monetise devaluation when public disclosure weakens the human alternative before licensing. Under restricted pre-disclosure contracting, demonstration can create the market for the machine, make private development exceed its social value, and reward both technical quality and pure salience. Complete contracting over secrecy can overturn the public-demonstration prediction for a single affected firm, while dispersed incidence and free riding restore a role for public disclosure. The strongest result, that formerly meaningful tasks become better targets than neutral tasks, additionally requires retained human performance to become a disamenity; the broader devaluation and development results do not.

In occupational and task-portfolio settings, employment alone is therefore an incomplete measure of technological change. The relevant objects include which tasks remain, how salient the fallback is, how wages and rents adjust, and what workers understand their continued performance to contribute. Higher income is not by itself a theory of human purpose. Ikigai, on Stevenson (2026)'s telling, was never confined to the job; the question her argument leaves open is how much of it currently resides there. The narrower point established here is that whatever meaning work does supply can begin to leak away while workers remain employed, paid, and productive—the machine need never be used. Whether it has begun to do so is, ultimately, an empirical question.

Appendices

A Proofs

Proof of Proposition 1. Equation (4) gives $m(1, a) - m(0, a) = -\beta\chi(a) < 0$. Equation (5) then gives the wage change $w^R(1, a) - w^R(0, a) = \theta\beta\chi(a) > 0$, and (6) gives the surplus change $S_H(1; a) - S_H(0; a) = -\theta\beta\chi(a) < 0$. Initially $(x, s) = (0, 0)$, automation is infeasible, and Assumption 3 gives $S_H(0; a) > 0$. After full salience $(x, s) = (1, 1)$, the maintained condition makes human production strictly better than automation and non-production in (7). $\square$

Proof of Proposition 2. Before the shock, worker utility is $w_0 - c - e + \theta m(0, a) = \omega$. After full salience and the wage adjustment, it is $w_1 - c - e + \theta m(1, a) = \omega + \rho L(a) - L(a) = \omega - (1 - \rho)L(a)$, so the worker remains willing to perform the task when $\omega \geq (1 - \rho)L(a)$ and worker utility falls by $(1 - \rho)L(a)$. Firm profit from human production changes from $y - w_0 = S_H(0; a) - \omega$ to $y - w_1 = S_H(0; a) - \omega - \rho L(a)$, a fall of $\rho L(a)$. Adding the worker and firm changes cancels the wage transfer, so total surplus falls by $L(a)$. Automation is strictly selected after salience exactly when it exceeds post-adjustment human profit and zero; it would not have strictly displaced the pre-salience contract exactly when $A \leq S_H(0; a) - \omega$. Combining the inequalities gives (10). $\square$

Proof of Proposition 3. At $(x, s) = (1, 1)$, automation is strictly adopted if and only if $A > \max\{S_H(1; a), 0\}$. If human meaning remained at $s = 0$, automation would not strictly dominate human production if and only if $A \leq S_H(0; a)$. This gives (11). For the deployment-cost restatement in the accompanying footnote, substitute $A = a - k$ and impose $k \geq 0$: writing $B = \max\{S_H(1; a), 0\}$, the restrictions are $k < a - B$, $k \geq a - S_H(0; a)$, and $k \geq 0$, an interval of length $a - B - \max\{0, a - S_H(0; a)\} = \min\{a, S_H(0; a)\} - B$. If $S_H(1; a) \geq 0$ and $a \geq S_H(0; a)$, this equals $S_H(0; a) - S_H(1; a) = L(a)$. $\square$

Proof of Corollary 1. Automation strictly displaces the formerly meaningful task if and only if $A > \max\{S_H(1; a), 0\}$, and does not strictly displace the neutral task if and only if $A \leq S_N$. The interval is non-empty exactly when $\max\{S_H(1; a), 0\} < S_N$. Since $S_N > 0$, this is equivalent to $S_H(1; a) < S_N$. By (12), $S_H(1; a) = S_N + \theta m(1, a)$, so the condition is $m(1, a) < 0$. $\square$

Proof of Proposition 4. Assumption 3 gives $S_H(0; a) > 0$, and (9) gives $S_H(0; a) > S_H(1; a)$; hence $S_H(0; a) > \max\{S_H(1; a), 0\}$. Since $u \mapsto [A - u]_+$ is weakly decreasing, $\bar{p}_M^D \geq \bar{p}_M^P$. If $A \leq \max\{S_H(1; a), 0\}$, both caps are zero. If $A > \max\{S_H(1; a), 0\}$, the public cap is

positive and strictly exceeds the private cap, whether $A \leq S_H(0; a)$ or $A > S_H(0; a)$. This proves parts (i) and (ii). For part (iii), if $A \leq S_H(0; a)$, the private cap is zero and neither private profitability nor a strict welfare gain is possible for $R \geq 0$. If $A > S_H(0; a)$, then $\bar{p}_M^P = A - S_H(0; a)$, so $\bar{p}_M^P > R$ if and only if $A - R > S_H(0; a)$, which is exactly the welfare comparison under Assumptions 1 and 4. $\square$

Proof of Proposition 5. By (12), $S_H(1; a) = S_N + \theta m(1, a)$. If $m(1, a) < 0$, then $\max\{S_H(1; a), 0\} < S_N$, so the public meaningful-task cap weakly exceeds the neutral cap and is strictly larger whenever $A > \max\{S_H(1; a), 0\}$; the stated development-cost interval follows. If $m(1, a) = 0$, the relevant outside options and fee caps coincide. If $m(1, a) > 0$, then $S_H(1; a) > S_N > 0$, reversing the weak ranking, with strict inequality when $A > S_N$. $\square$

Proof of Corollary 2. Under the maintained adoption condition, $\bar{p}_M^D = A - \max\{S_H(1; a), 0\}$, so development is profitable exactly when $R < A - \max\{S_H(1; a), 0\}$. Following development and adoption, welfare is $A - R$, compared with $S_H(0; a)$ before development, so welfare falls exactly when $R > A - S_H(0; a)$. Combining the inequalities with $R \geq 0$ gives (17). Non-emptiness follows from $S_H(0; a) > \max\{S_H(1; a), 0\}$: if $A < S_H(0; a)$, $R = 0$ belongs to the set, and if $A \geq S_H(0; a)$, the interval $(A - S_H(0; a), A - \max\{S_H(1; a), 0\})$ is non-empty. $\square$

Proof of Proposition 6. Holding k fixed gives $\partial A / \partial a = 1$. From (6),

$$\frac{\partial S_H(s; a)}{\partial a} = -s\theta\beta\chi'(a), \quad \frac{\partial S_H(s; a)}{\partial s} = -\theta\beta\chi(a).$$

The strict adoption condition and the sign of $S_H(s; a)$ are locally preserved. Since $S_H(s; a) > 0$, the meaningful-task cap is $\bar{p}_M = A - S_H(s; a)$, with quality derivative $1 + s\theta\beta\chi'(a)$ and salience derivative $\theta\beta\chi(a)$. The neutral cap, where positive, is $A - S_N$, with quality derivative one and salience derivative zero, and conditional on adoption social surplus is $A - R$, also with quality derivative one and salience derivative zero. If instead $S_H(s; a) < 0$, the cap is A , which gives the footnoted case. $\square$

Proof of Proposition 7. Before the capability, $(x, s) = (0, 0)$ and welfare is $S_H(0; a) > 0$. After full salience, welfare is $\max\{S_H(1; a), A, 0\}$. Substituting each maximising alternative gives the three stated changes; at ties the alternatives have the same welfare. $\square$

Proof of Proposition 8. When $m_E^0 > m_P$, the exposed-minus-protected payoff difference is strictly increasing in θ , so exposed work is chosen by an upper tail and market clearing gives $F(\theta_0^*) = n_P$. When $m_P > m_E^1$, the difference is strictly decreasing in θ , so exposed

work is chosen by a lower tail and market clearing gives $F(\theta_1^*) = n_E$. The stated sets follow from the two cutoff rules. $\square$

B Economically Viable Replaceability

The baseline takes the view that a technically credible capability can improve the perceived fallback even when deployment remains expensive. This appendix considers the narrower case in which the worker responds only to an economically valuable substitute. Replace (2) with $b^{\text{ev}}(s, a, k) = b_0 + s\chi((a - k)_+)$, where $(u)_+ \equiv \max\{u, 0\}$, and define

$$m^{\text{ev}}(s, a, k) = \alpha v + \beta [\Delta_0 - s\chi(A_+)], \quad A_+ \equiv \max\{A, 0\}. \quad (22)$$

Under full salience, post-capability human surplus is

$$S_H^{\text{ev}}(1; A) = S_H(0; a) - \theta\beta\chi(A_+). \quad (23)$$

Proposition 9. *Suppose Assumptions 2 and 3 hold, and suppose $\chi : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is continuous, satisfies $\chi(0) = 0$, and is strictly increasing. Under (22):*

(i) *The full-salience meaning loss is*

$$m^{\text{ev}}(1, a, k) - m^{\text{ev}}(0, a, k) = -\beta\chi(A_+).$$

It is zero when $A \leq 0$ and strictly negative when $A > 0$.

(ii) *Meaning-induced adoption occurs if and only if*

$$0 < A \leq S_H(0; a) \quad \text{and} \quad A + \theta\beta\chi(A) > S_H(0; a). \quad (24)$$

(iii) *There is a unique threshold $A^* \in (0, S_H(0; a))$ satisfying*

$$A^* + \theta\beta\chi(A^*) = S_H(0; a). \quad (25)$$

Meaning-induced adoption occurs exactly for $A \in (A^, S_H(0; a)]$. For fixed a , the corresponding deployment costs are*

$$k \in [\max\{0, a - S_H(0; a)\}, a - A^*) \quad (26)$$

whenever $a > A^$, and the set is empty otherwise.*

Proof. Equation (22) gives the first difference directly. Strict monotonicity and $\chi(0) = 0$ establish part (i). Meaning-induced adoption requires $A > \max\{S_H^{\text{ev}}(1; A), 0\}$ and $A \leq S_H(0; a)$. The first inequality implies $A > 0$, so substituting (23) gives (24). For part (iii), define $h(A) = A + \theta\beta\chi(A) - S_H(0; a)$ on $[0, S_H(0; a)]$. It is continuous and strictly increasing, with $h(0) < 0$ and $h(S_H(0; a)) > 0$, so the intermediate value theorem yields the unique A^* . Substituting $A = a - k$ and imposing $k \geq 0$ gives (26). $\square$

The conservative specification makes the meaning effect economic rather than purely technical. The retained-work comparative static survives for every positive-net-value substitute, but the adoption region is governed by an implicit threshold.

C Task-Portfolio Decomposition

This appendix states the accounting identity discussed in Section 5. An occupation consists of a continuum of potential tasks $j \in [0, 1]$. At date $t \in \{0, 1\}$, labour performs the measurable set $H_t \subseteq [0, 1]$ with intensity or time weight $\ell_t(j) \geq 0$, where $\ell_t(j) = 0$ for $j \notin H_t$ and total time is normalised by $\int_0^1 \ell_t(j) dj = 1$. Task j has perceived social value $v_j > 0$, initial fallback b_{j0} , quality a_j , objective feasibility $x_j(t) \in \{0, 1\}$, and salience $s_j(t) \in [0, 1]$, with $s_j(t) = 0$ whenever $x_j(t) = 0$. If task j remains with labour, it supplies meaning $m_j(t) = \alpha v_j + \beta[v_j - b_{j0} - s_j(t)\chi_j(a_j)]$, exactly as in the baseline. With a non-meaning burden $e_j(t) \geq 0$, the net non-pecuniary task value for a worker with taste θ is $q_j(t; \theta) = \theta m_j(t) - e_j(t)$, and occupation-level non-pecuniary value is

$$Q_t(\theta) = \int_0^1 \ell_t(j) q_j(t; \theta) dj, \quad (27)$$

assuming the integrand is integrable. The opportunity cost of time is omitted because total working time is held fixed. The allocation sets H_t and intensities ℓ_t are reduced-form objects; the decomposition does not determine which tasks firms automate or how intensities are chosen.

Proposition 10. $Q_1(\theta) - Q_0(\theta)$ equals a retained-task and reweighting term, $\int_{H_0 \cap H_1} [\ell_1(j) q_j(1; \theta) - \ell_0(j) q_j(0; \theta)] dj$, minus a removed-task term, $\int_{H_0 \setminus H_1} \ell_0(j) q_j(0; \theta) dj$, plus a new-task term, $\int_{H_1 \setminus H_0} \ell_1(j) q_j(1; \theta) dj$.

Proof. Because $\ell_t = 0$ outside H_t , split each integral in (27) over $H_0 \cap H_1$, $H_0 \setminus H_1$, and $H_1 \setminus H_0$ and collect terms. $\square$

The retained-task and reweighting term records tasks that stay with labour, including the baseline devaluation mechanism; the removed-task term records the weighted net value of tasks no longer performed by labour, which raises the occupation's value when the removed task was sufficiently burdensome that $q_j(0; \theta) < 0$; and the new-task term records value from newly created human tasks.

D Augmentation, Technology Design, and Integrated Development

D.1 Productive augmentation with lower meaning

This appendix formalises the augmentation comparison discussed in Section 5. Let $\tau \in \{0, 1\}$ index whether a worker uses an augmentation technology. The task remains assigned to labour under either value. Let y_τ denote market output, $v_\tau > 0$ perceived social value, $e_\tau \geq 0$ direct burden, and $r_\tau \in \mathbb{R}$ a signed attribution coefficient. Meaning is

$$m_\tau = \underbrace{\alpha v_\tau}_{\text{useful-output effect}} + \underbrace{\beta r_\tau v_\tau}_{\text{attributable-contribution effect}}. \quad (28)$$

The mapping to the baseline is

$$r_\tau = \frac{v_\tau - b_\tau}{v_\tau}.$$

The conventional responsibility-share case has $r_\tau \in [0, 1]$. Allowing $r_\tau < 0$ preserves the baseline region in which personal performance is understood as inferior to the available fallback. A nonnegative responsibility share is therefore a conservative special case rather than a different unexplained microfoundation.

Assume $y_1 > y_0$ and $v_1 \geq v_0$: augmentation increases productivity and does not reduce perceived usefulness.

Observation 1. *Augmentation reduces meaning if and only if*

$$\alpha(v_1 - v_0) + \beta(r_1 v_1 - r_0 v_0) < 0. \quad (29)$$

Proof. Subtract m_0 from m_1 in (28). □

A decision-support tool can increase output and preserve responsibility. A system that generates an answer for formal approval can raise output while lowering the value attrib-

uted to human judgment. The retained worker may therefore become more productive and less fulfilled.

D.2 Technology choice

Suppose technology τ costs C_τ and the firm recruits a worker at the reservation wage. Its surplus is

$$S_\tau = y_\tau - C_\tau - c - e_\tau + \theta m_\tau. \quad (30)$$

Observation 2. *Technology 1 is weakly preferred to technology 0 if and only if*

$$(y_1 - y_0) - (C_1 - C_0) - (e_1 - e_0) + \theta(m_1 - m_0) \geq 0. \quad (31)$$

If $e_1 = e_0$ and technology 1 has lower net production value, it can nevertheless be selected whenever

$$\theta(m_1 - m_0) \geq (y_0 - C_0) - (y_1 - C_1). \quad (32)$$

Proof. Substitute (30) into $S_1 \geq S_0$, cancel c , and rearrange. □

A technology can therefore be attractive because it raises output, reduces deployment cost, removes an ordinary burden, or preserves meaning. Under full wage adjustment, the firm internalises meaning through the compensation required to recruit labour.

D.3 Integrated development benchmark

Suppose the employing firm pays R to create the capability for its own use. Let $s_I \in [0, 1]$ denote how salient the internal capability becomes to its worker. After development, the firm's best production value is

$$\max\{A, S_H(s_I; a), 0\} - R, \quad (33)$$

which it compares with pre-development human production $S_H(0; a)$. If the firm automates, its return is $A - S_H(0; a) - R$, independent of s_I . For the otherwise identical neutral task the corresponding return is $A - S_N - R$; since $S_H(0; a) > S_N$, the neutral task is the more attractive internal automation target.

If the firm retains human production, its return is

$$S_H(s_I; a) - S_H(0; a) - R = -s_I L(a) - R.$$

This return is strictly negative whenever $s_I > 0$, and equals $-R$ if the internal capability remains entirely non-salient. Thus an integrated firm does not profit merely from weakening an amenity that lowered its own wage bill. This internalisation is complete only under full wage adjustment: with pass-through $\rho < 1$, a salient internal capability costs the firm $\rho L(a)$ while its retained worker bears the remainder, so integration aligns private and social incentives exactly only in the posted-wage benchmark. The fully public benchmark corresponds to $s_I = 1$; a confidential internal capability corresponds to $s_I = 0$. Under Assumptions 1 and 4, a social planner uses the same comparison.

E Robustness of the Meaning and Welfare Specifications

E.1 General and bounded attributable contribution

Replace the linear attributable-contribution term with a weakly increasing function $g : \mathbb{R} \rightarrow \mathbb{R}$:

$$m_g(s, a) = \alpha v + \beta g(\Delta_0 - s\chi(a)). \quad (34)$$

Define

$$S_{Hg}(s; a) = y - c - e + \theta m_g(s, a) \quad (35)$$

and

$$L_g(a) = S_{Hg}(0; a) - S_{Hg}(1; a) = \theta \beta [g(\Delta_0) - g(\Delta_0 - \chi(a))]. \quad (36)$$

Maintain Assumption 2, suppose g is weakly increasing with $g(\Delta_0) > g(\Delta_0 - \chi(a))$, and suppose the analogues of paid and viable initial human work hold: $c + e > \theta m_g(0, a)$ and $S_{Hg}(0; a) > 0$.

Proposition 11. (i) $L_g(a) > 0$, and Propositions 1 and 3 continue to hold with S_{Hg} and L_g in place of S_H and L . (ii) Under Assumption 5, the public fee cap exceeds the private cap whenever positive, and the meaning-externality term $S_{Hg}(0; a) - \max\{S_{Hg}(1; a), 0\}$ is strictly positive, so the development wedge does not require negative meaning. (iii) If $S_N > 0$, the reversal relative to a neutral task, and the associated targeting reversal, occur if and only if $m_g(1, a) < 0$; for the bounded specification $g(u) = u_+$, $m_g(s, a) \geq 0$ while $L_g(a) = \theta \beta \min\{\chi(a), \Delta_0\} > 0$, so devaluation survives but amenity reversal is impossible.

Proof. Monotonicity and $g(\Delta_0) > g(\Delta_0 - \chi(a))$ imply $L_g(a) > 0$. The proofs of the retained-work and adoption results use the strict ordering $S_{Hg}(0; a) > S_{Hg}(1; a)$, together with initial viability, so their wage, surplus, interval, and fee-cap arguments carry over.

Under Assumption 5, the same ordering implies that the public outside option is strictly below the private one whenever the public cap is positive.

Since the neutral task has surplus S_N , we have $S_{Hg}(1; a) = S_N + \theta m_g(1, a)$. Given $S_N > 0$, the reversal comparison $\max\{S_{Hg}(1; a), 0\} < S_N$ holds if and only if $S_{Hg}(1; a) < S_N$, which holds if and only if $m_g(1, a) < 0$. If $g(u) = u_+$, then $m_g(s, a) = \alpha v + \beta[\Delta_0 - s\chi(a)]_+ \geq 0$, while (36) becomes the stated positive minimum. $\square$

The bounded case is useful for interpretation. If a worker experiences the loss of attributable contribution only as the disappearance of a positive amenity, the basic pre-displacement devaluation remains, as do its adoption and external-development implications. What disappears is the stronger claim that continued human performance becomes worse to staff than an otherwise identical neutral task.

E.2 Separate private and social output values

Let z index technical quality and let $m(s, z)$ be defined as in (4), with $\chi(z)$. Let y^p and y^s denote private and social values of human output, and let $a^p(z)$ and $a^s(z)$ denote private and social values of automated output. Define

$$S_H^p(s; z) = y^p - c - e + \theta m(s, z), \quad S_H^s(s; z) = y^s - c - e + \theta m(s, z),$$

$$A^p(z) = a^p(z) - k, \quad A^s(z) = a^s(z) - k.$$

The firm chooses among $S_H^p(s; z)$, $x A^p(z)$, and zero, whereas the planner compares $S_H^s(s; z)$, $x A^s(z)$, and zero. If a public demonstration supports a positive licence fee and induces adoption, the developer's public licensing margin minus the gross social gain is

$$\begin{aligned} & [A^p(z) - \max\{S_H^p(1; z), 0\}] - [A^s(z) - S_H^s(0; z)] \\ &= \underbrace{S_H^p(0; z) - \max\{S_H^p(1; z), 0\}}_{\text{meaning externality in the buyer's private outside option}} + \underbrace{[A^p(z) - S_H^p(0; z)] - [A^s(z) - S_H^s(0; z)]}_{\text{ordinary product-market wedge}}. \end{aligned} \quad (37)$$

Assumption 1 sets the second term to zero. Relaxing it does not eliminate the meaning externality; it adds a familiar output-side wedge whose sign is independent.

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